O_X-MODULES ON THE SMALL ETALE SITE AS COMPATIBLE FAMILIES OF MODULES ON THE PIECES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 03LV: sources/stacks-chow/tags.txt, source lines 4727–4727. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03LS: sources/stacks-chow/tags.txt, source lines 4724–4724. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. A sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules is given by the following data:*

(1) *for every $U \in \mathrm{Ob}(X_{étale})$ a sheaf $\mathcal{F}_U$ of $\mathcal{O}_U$-modules on $U_{étale}$,*
(2) *for every $f : U' \to U$ in $X_{étale}$ an isomorphism $c_f : f^*_{small}\mathcal{F}_U \to \mathcal{F}_{U'}$.*

These data are subject to the condition that given any $f : U' \to U$ and $g : U'' \to U'$ in $X_{étale}$ the composition $c_g \circ g^*_{small} c_f$ is equal to $c_{f \circ g}$.

Proof. Combine Lemmas 03LV and 03LS, and use the fact that any morphism between objects of $X_{étale}$ is an étale morphism of schemes. □

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.